MoCafe v2.00 — A Monte-Carlo Dust Radiative-Transfer Code with Scattering and Thermal Emission: User Guide

Abstract

MoCafe is a parallel (Fortran 90 + MPI) Monte-Carlo dust radiative-transfer code. Version 2.00 extends the scattering-only v1.20 into a panchromatic code that computes *dust thermal emission*: it transports photon packets over a wavelength grid, tallies the radiation field in every cell, and re-emits the absorbed energy as thermal dust emission through either the Lucy (1999) method coupled to the SEDUST grain library (equilibrium and stochastically heated grains, including PAH features) or the Bjorkman & Wood (2001) immediate-reemission method. It also adds multiple stellar populations, an all-sky interior observer (for the Milky-Way case), and a Modified Random Walk for very optically thick regions. This guide covers installation, execution, the complete input reference, the dust-emission workflow, the output products, worked examples, and the underlying algorithms.

Contents

1. Overview

MoCafe follows photon packets emitted by one or more sources through a dusty medium. Packets scatter off dust grains (Henyey–Greenstein phase function or a tabulated Mueller matrix) and, in v2.00, are absorbed and re-emitted as thermal radiation. At every interaction a contribution is "peeled off" toward each observer to build an image or spectral-energy distribution (SED).

What is new in v2.00.

- **Multi-wavelength (SED) transport** (`par%use_sed`): one run covers the full UV-to-far-IR range on a log-spaced wavelength grid.
- **Per-cell mean intensity** $J_\lambda(x, y, z)$ (`par%save_jlam`) via the Lucy (1999) pathlength estimator.
- **Dust thermal emission** (`par%use_dustemis`) in two modes: *Mode 1* (Lucy + SEDUST) and *Mode 2* (Bjorkman & Wood 2001).
- **Multiple stellar populations** (`par%nsource`) with independent spectra, luminosities, and spatial distributions.
- **All-sky interior observer** (`par%allsky`) for the Milky-Way case.
- **Modified Random Walk** (`par%use_mrw`) plus Russian roulette for very optically thick media.

Everything above is opt-in; a plain monochromatic scattering run behaves exactly as in v1.20. The design and validation record is in `MoCafe_v2.00_PLAN.md`.

Units. In scattering-only mode MoCafe is unit-agnostic and the image is in `(luminosity unit)` $\mathrm{cm^{-2}\,sr^{-1}}$. In SED / emission mode a physical length scale is required (through `par%distance_unit`: one grid length unit equals one distance unit) and the source luminosity (`par%luminosity`) is in $\mathrm{erg\,s^{-1}}$, so intensities and dust temperatures are absolute.

2. Installation

2.1 Prerequisites

- A Fortran 2008 MPI compiler: Intel oneAPI (`mpiifort` or `mpiifx`) or GNU (`mpif90`). MoCafe is MPI-only.
- **CFITSIO** (required).
- **HDF5** 1.14+ with Fortran bindings built by the same compiler (optional; on by default).
- Python 3 with `numpy`, `astropy`, `h5py` (and `scipy` for the Illustris converter) for the analysis tools.

2.2 Building MoCafe

```
make                              # HDF5=1 (default) -> MoCafe.x
make HDF5=0                       # CFITSIO only
make HDF5_PREFIX=/usr/local/hdf5  # override HDF5 location
make F90=mpif90                   # force a compiler
make DEBUG=1                      # checked / traceback flags
```

The compiler is auto-selected `mpiifort` $\rightarrow$ `mpiifx` $\rightarrow$ `mpif90`. The build always passes `-cpp -DMPI` and (by default) `-DHDF5`. When you add a source file, append its object to `OBJS` in the `Makefile` in dependency order.

2.3 Building the SEDUST library

The Lucy emission engine (Mode 1) links the vendored SEDUST library, kept self-contained under `sedust/`. Build it once and populate its optics tables:

```
cd sedust/sed && ./build_lib.sh    # -> sedust/sed/lib/libsedust.a  (Intel)
./populate_data.sh                 # copy the ~55 MB optics tables into sedust/
```

The MoCafe `Makefile` links `sedust/sed/lib/libsedust.a` automatically (variable `SEDUST_LIBDIR`). The SEDUST module named `mathlib` is vendored as `sed_mathlib` to avoid a clash with MoCafe's own `mathlib`. The bulky optics data are not tracked in git; `populate_data.sh` copies them from the canonical SEDUST tree. Mode 2 (Bjorkman & Wood) needs only the extinction table and not SEDUST.

3. Running MoCafe

```
mpirun -np <N> ./MoCafe.x <input>.in
```

The single argument is a Fortran namelist file. Every tunable quantity is a field of the global `par` structure and is set as `par%<name> = <value>` inside a `&parameters / . . . /` block. Sample inputs and `run.sh` scripts live under `examples/`. The default of every parameter is the initializer of `params_type` in `src/define.f90`.

Reproducibility. A run is bit-identical when `par%iseed` is a fixed non-zero value, `par%use_master_slave` is `.false.` (equal-share), and the number of MPI ranks is fixed (`par%iseed` = 0 seeds from the clock/PID).

4. Input Parameter Reference

Parameters are grouped below. Types: `L` logical, `I` integer, `R` real, `S` string, `[ ]` array.

4.1 Simulation control

Parameter	Default	Description
`no_photons`	1e5	Number of photon packets
`no_print`	1e7	Progress-print interval (photons)
`iseed`	0	RNG seed (0 $\rightarrow$ seeded from clock/PID)
`use_master_slave`	.true.	MPI mode: master–slave vs. equal-share
`num_send_at_once`	10000	Batch size (master–slave)
`luminosity`	1.0	Total source luminosity [erg/s] (SED/emission mode)

4.2 Grid geometry (Cartesian)

Parameter	Default	Description
`nx, ny, nz`	1,1,11	Grid resolution
`xmax, ymax, zmax`	1.0	Half-sizes of the box (grid spans $[-\max, \max]$)
`rmax`	-999	If > 0 and not a cylinder, forces a cube of side $r_{\max}$ (sphere)
`nr`	-999	If > 1, sets nx=ny=nz=nr
`geometry`	' '	”, ’sphere’, ’cylinder’, or ’Plummer’ (auto-set from source)
`xyz_symmetry`	.false.	Exploit octant symmetry (disabled if any $n = 1$)
`z_symmetry`	.false.	Exploit reflection about $z = 0$
`xy_periodic`	.false.	Periodic in x, y

4.3 Grid type and media

Parameter	Default	Description
`grid_type`	’car’	’car’ (Cartesian), ’clump’ (clumpy), ’amr’ (octree)
`amr_file`	' '	Generic AMR octree file (FITS/HDF5/text) for ’amr’
`dust_model`	’global_dgr’	AMR leaf opacity: ’global_dgr’, ’from_file’, ’laursen09’
`DGR, Z_global, Z_ref, f_ion_dust`	—	Dust-to-gas / metallicity parameters (AMR)
`clump_radius`	-1	Clump radius (clumpy medium)
`clump_f_cov / _f_vol / _N_clumps`	-1	Clump count specification
`clump_tau0 / _ndust / _nH`	-1	Per-clump opacity specification

The clumpy and AMR media are documented in `docs/MoCafe_clump.pdf` and `docs/MoCafe_amr.pdf`. The full dust-emission suite runs on both the Cartesian grid and the AMR octree (§5.8); the clumpy medium is scattering-only.

4.4 Distance unit

Parameter	Default	Description
`distance_unit`	'kpc'	", 'pc', 'au', 'kpc'; sets `distance2cm`. In SED mode one grid unit = one distance unit (physical scale)

4.5 Source geometry (single source)

Parameter	Default	Description
`source_geometry`	'point'	'point', 'uniform', 'uniform_xy', 'gaussian', 'exponential', 'external_{sph,cyl,rec}'
`xs_point, ys_point, zs_point`	0.0	Point-source location
`source_zscale`	NaN	Scale height for 'gaussian'/'exponential'
`radiation_angular_PDF_file`	' '	Angular PDF for external illumination

Multiple sources are described in §5.5.

4.6 Dust and scattering (grey / monochromatic)

Parameter	Default	Description
`albedo`	0.3253	Single-scattering albedo
`hgg`	0.6761	Henyey–Greenstein asymmetry g
`cext_dust`	1.61e-21	Dust extinction per H nucleus [cm^2]
`scatt_mat_file`	' '	Mueller-matrix table (`data/mueller_*.dat`); enables Stokes
`use_stokes`	.true.	Stokes polarization (auto-off without a matrix file)

In SED mode the wavelength-dependent albedo, g, and C_{ext} are read from `par%kext_file` and override these scalars (§5.1).

4.7 Optical depth and density

Parameter	Default	Description
`taumax`	-999	Radial optical depth, center $\rightarrow$ edge (sets the density scale)
`tauhomo`	-999	Optical depth of the volume-equivalent homogeneous medium
`density_file`	' '	3-D density cube (FITS/HDF5)
`reduce_factor`	1	Density-file down-sampling

Parameter	Default	Description
`centering`	0	Density-file re-centering mode
`density_zscale`, `density_rscale`, `density_powerlaw_index`	NaN	Analytic density modulations

In SED mode `par%taumax`/`par%tauhomo` are defined at the reference wavelength `par%lambda_ref`.

4.8 Peel-off observers

Parameter	Default	Description
`nxim, nyim`	129	Image pixels
`dxim, dyim`	NaN	Pixel size [deg] (auto to cover the grid if unset)
`distance`	NaN	Observer distance; renormalizes (obsx,obsy,obsz)
`inclination_angle`, `position_angle, phase_angle`	NaN	Observer angles [deg] (arrays)
`alpha, beta, gamma`	NaN	Euler angles [deg] (alternative)
`obsx, obsy, obsz`	NaN	Explicit observer direction (alternative)

Up to `MAX_OBSERVERS = 181` observers per run. Angle conventions are in `docs/MoCafe_Geometry.pdf`.

4.9 Output control

Parameter	Default	Description
`file_format`	'hdf5'	'hdf5' or 'fits'; authoritative for the container
`out_file`	''	Output base name (from the input name if empty)
`out_bitpix`	-32	FITS bitpix (–32 float32, –64 float64)
`save_direc0`	.false.	Also write the unattenuated direct image
`sightline_tau`	.false.	Also write a per-pixel sight-line optical-depth map

5. Dust Thermal Emission

The emission workflow is: *SED transport* (panchromatic scattering) $\rightarrow$ J_λ *tally* $\rightarrow$ *dust emission* $\rightarrow$ observer SEDs. Enable it with `par%use_sed`, and add `par%save_jlam` and `par%use_dustemis` for the emission.

5.1 Multi-wavelength (SED) mode

Parameter	Default	Description
use_sed	.false.	Enable panchromatic transport
nlambda	128	Number of log-spaced wavelength bins
lambda_min, lambda_max	0.0912, 2000	Wavelength range [μm]
lambda_ref	0.55	Reference wavelength [μm] for the grid opacity and τ
kext_file	' '	Table λ, ϖ, $\langle\cos\rangle$, $C_{\rm ext}/{\rm H}$
source_spectrum	' '	Two-column source spectrum λ, L_λ
tstar	-999	Planck source temperature [K] (if no spectrum file)

Wavelength-dependent extinction is applied as a per-photon grey rescaling

$$s_{\rm ext}(\lambda) = \frac{C_{\rm ext}(\lambda)}{C_{\rm ext}(\lambda_{\rm ref})}, \tag{1}$$

so the grid stores only the reference-wavelength opacity and every optical depth is multiplied by $s_{\rm ext}$ (the same device as the Jonsson 2006 τ scan). All three ray tracers (Cartesian, clumpy, AMR) are therefore shared with the monochromatic path, and a run with $s_{\rm ext} = 1$ is bit-identical to the mono code.
Restrictions: SED mode is incompatible with Stokes polarization and with the $(a, g)/\tau$ single-run scans, and (in this version) supports internal sources only.

5.2 Per-cell mean intensity J_λ

Parameter	Default	Description
save_jlam	.false.	Tally J_λ and write <base>_jlam

The mean intensity is accumulated with the Lucy (1999) pathlength estimator: each path segment of length $d\ell$ contributes $w L_{\rm pk}\, d\ell$ to its (wavelength-bin, cell) slot, and

$$J_\lambda({\rm cell}) = \frac{1}{4\pi V_{\rm cell}\, \Delta\lambda\, d_{\rm cm}^2} \sum_{\rm paths} w L_{\rm pk}\, d\ell, \tag{2}$$

where $L_{\rm pk}$ is the packet luminosity [erg/s] and $d_{\rm cm}$ the distance-unit conversion. The forced first-flight is tallied analytically (a zero-variance direct term). A built-in energy check reports the absorbed luminosity from the tally against an independent event counter. Memory is $\approx n_\lambda \times n_{\rm cell} \times 8$ bytes per MPI rank ($n_{\rm cell} = n_x n_y n_z$ on the Cartesian grid, or the number of leaves on the AMR octree; see §5.8).

5.3 Mode 1: Lucy (1999) with SEDUST

Parameter	Default	Description
use_dustemis	.false.	Compute dust emission (requires par%use_sed)

Parameter	Default	Description
`dust_emission_method`	'lucy'	'lucy' (SEDUST) or 'bw01' (Bjorkman & Wood)
`dust_model_sed`	'astrodust'	SEDUST model
`sed_qtable, sed_sizedist`	(vendored)	SEDUST optics / size-distribution paths
`sed_workdir`	(vendored)	Directory SEDUST reads its `../data/` tables from
`sed_NT, sed_Tlo, sed_Thi`	200, 2.7, 5000	Grain temperature grid
`dust_niter`	1	Lucy iterations (dust self-absorption)
`dust_no_photons`	1e6	Dust-emission photons per iteration
`dust_tol`	1e-3	Convergence on the total emitted luminosity

The `'lucy'` method requires `par%save_jlam`. For each cell the local mean intensity J_λ (converted to SI and resampled onto the SEDUST wavelength grid) is passed to SEDUST, which returns the emission spectrum from the full grain-size distribution — equilibrium grains plus stochastically heated small grains and PAH features. The *shape* is taken from SEDUST while the per-cell bolometric luminosity is set to the locally absorbed power,

$$L_{\rm emit}({\rm cell}) \;=\; L_{\rm abs}({\rm cell}) \;=\; \rho\kappa \sum_{\lambda} s_{\rm ext}(\lambda)\,[1-\varpi(\lambda)] \left(\textstyle\sum_{\rm paths} w\, L_{\rm pk}\, d\ell \right)_{\lambda,{\rm cell}}, \tag{3}$$

so total energy is conserved by construction (radiative equilibrium). With `par%dust_niter` > 1 the re-emitted photons are transported so that dust self-absorption heats other cells; the iteration converges when the total emitted luminosity stops changing (needed at high infrared optical depth).

Emission-solver speed options. The exact stochastic solve costs ~ 0.1 s per cell; two faster paths are available:

Parameter	Default	Description
`dust_single_teq`	.false.	One mixture equilibrium T per cell; fastest ($\sim$55$\times$); true $T_{\rm eq}$; *no PAH/stochastic features*
`dust_fast_table`	.false.	Interpolate emission vs. field intensity U over a fixed reference shape; $\sim$1% SED-shape error; keeps stochastic/PAH
`dust_nU`	40	U-grid size for the table path

5.4 Mode 2: Bjorkman & Wood (2001)

Set `par%dust_emission_method` = `'bw01'`. This is an approximate, equilibrium, mixture mean-opacity method with *no iteration*: fixed-energy packets random-walk through scatter and absorb+immediate-reemit events until they escape. Each absorption raises the local temperature (from the running absorbed power) and the packet wavelength is resampled from the temperature-correction spectrum $\propto \kappa_{\rm abs}\,\partial B_\lambda/\partial T$, so the time-averaged emission equals $\kappa_{\rm abs} B_\lambda(T)$ without iterating. Mode 2 needs only `par%kext_file` (not SEDUST) and does not reproduce PAH or stochastic-heating features. It writes `<base>_bwdust` (equilibrium $T_{\rm dust}$ and absorbed-power maps); the emergent dust emission is added to the observer `Scattered` SED image.

5.5 Multiple stellar populations

Parameter	Default	Description
`nsource`	1	Number of stellar components (> 1 activates multi-source)
`src_geometry(i)`	'point'	Per-source geometry: point/uniform/gaussian/exponential
`src_tstar(i)`	-999	Per-source Planck temperature [K]
`src_spectrum(i)`	' '	Per-source spectrum file (overrides src_tstar)
`src_lum(i)`	-999	Per-source luminosity [erg/s] (equal split if unset)
`src_x/y/z(i)`	0	Per-source position ('point')
`src_zscale(i)`	-999	Per-source scale height ('gaussian'/'exponential')

Sources are sampled in proportion to their luminosity, and `par%luminosity` becomes $\sum_i$`src_lum(i)`. A typical use is a hot young disk plus a cool old bulge (`examples/galaxy/`).

5.6 All-sky interior observer (Milky Way)

Parameter	Default	Description
`allsky`	.false.	Observer inside the medium; bin every event by sky direction
`allsky_x/y/z`	0	Interior observer position (code units)
`allsky_nlon, allsky_nlat`	360, 180	Equirectangular map size

Each scatter, dust-reemission, and direct event peels toward the interior observer and is binned by the sky direction it arrives from. The event-to-observer optical depth is integrated only over the segment to the observer. The output `<base>_allsky` is an equirectangular (ℓ, b, λ) surface-brightness cube — the diffuse Galactic light and far-infrared cirrus as seen from inside the disk (`examples/milkyway/`).

5.7 Modified Random Walk (high optical depth)

Parameter	Default	Description
`use_mrw`	.false.	Diffusion step in scattering-thick cells
`mrw_gamma`	2.0	Trigger threshold on the nearest-wall scattering optical depth

In a cell whose nearest-wall *scattering* optical depth exceeds `par%mrw_gamma`, MoCafe replaces the many small steps with a single diffusion step to the inscribed-sphere surface (Min et al. 2009; Robitaille 2010): it samples the total path length inside the sphere from the diffusion first-passage distribution, deposits the absorption-weighted path into the J_λ

tally, and decays the weight by $\exp(-\alpha_{\rm abs}\, ct)$. The Lucy transport also applies Russian roulette so high-albedo, high-τ clouds do not diffuse indefinitely. MRW is exact in absorbing dust (where it triggers rarely); its speed-up is large only when individual cells are very thick ($\tau_{\rm cell} \gg 1$) and nearly pure-scattering.

5.8 Dust emission on the AMR octree

The entire dust-emission suite — SED transport, the J_λ tally, Mode 1 (Lucy, including the single-$T_{\rm eq}$ and table options), Mode 2 (Bjorkman & Wood), and the Modified Random Walk — runs on the AMR octree (`par%grid_type = 'amr'`) exactly as on the Cartesian grid. A "cell" is simply an octree leaf: the radiation field, absorbed power, and dust temperature are tallied per leaf (per-leaf code-volume and opacity), and MRW uses the nearest leaf-face distance for its trigger. Set up the octree as usual (`par%grid_type = 'amr'`, `par%amr_file`, and `par%dust_model` for the leaf opacity) and add the emission parameters above. The derived files carry per-leaf arrays instead of Cartesian cubes: `<base>_jlam`, `<base>_dustsed`, and `<base>_bwdust` each add a `LeafXYZ` table of leaf-center coordinates so the per-leaf quantities can be placed in space. Example: `examples/amr_sphere/amr_dustemis.in` (a uniform level-5 octree sphere) reproduces the energy budget and SED of the equivalent Cartesian sphere.

5.9 A minimal emission run

```
&parameters
 par%no_photons  = 2.0e6
 par%use_sed     = .true.
 par%save_jlam   = .true.
 par%use_dustemis    = .true.
 par%dust_single_teq = .true.     ! fast equilibrium solve
 par%luminosity  = 3.828e33       ! erg/s
 par%nlambda     = 100
 par%lambda_min  = 0.0912
 par%lambda_max  = 2000.0
 par%kext_file   = 'data/kext_astrodust_MW.dat'
 par%tstar       = 1.0e4
 par%taumax      = 5.0
 par%source_geometry = 'point'
 par%distance_unit   = 'pc'
 par%rmax = 1.0
 par%nx = 33
 par%ny = 33
 par%nz = 33
 par%distance = 1.0e5
/
```

6. Output Products

`par%file_format` is authoritative: if `par%out_file`'s extension disagrees, MoCafe rewrites it (with a warning) so that the main file and all derived files share one container. HDF5 mirrors the FITS HDU layout (each image HDU becomes a group named after its `EXTNAME` with a `data` dataset; FITS keywords become group attributes).

Suffix	Written when	Contents
`_obs`	always	Observer images; in SED mode `Scattered`/`Direct` become (x, y, λ) cubes plus the wavelength / optics tables
`_jlam`	save_jlam	$J_\lambda(\lambda, x, y, z)$, $J_{\rm bol}$, energy-conservation keywords
`_dustsed`	use_dustemis ('lucy')	Emergent and intrinsic dust SED, a pixel-resolved `DustEmis_image`(x, y, λ), and per-cell $T_{\rm dust}/L_{\rm dust}$ maps
`_bwdust`	'bw01'	Equilibrium $T_{\rm dust}$ and absorbed-power maps
`_allsky`	allsky	Equirectangular (ℓ, b, λ) all-sky surface-brightness cube
`_obs_tau`	sightline_tau	Sight-line optical-depth map
`_stokes`	use_stokes	Stokes I, Q, U, V images

On the AMR octree the `_jlam`, `_dustsed`, and `_bwdust` images are per-leaf arrays (indexed by leaf) rather than Cartesian cubes, and each adds a `LeafXYZ` table of leaf-center coordinates. Read and convert the outputs with `python/mocafe_io.py` (`info`, `peek`, `convert`).

7. Worked Examples

Directory	Shows
`examples/point_source, uniform_source, external_*`	Cartesian grids and source geometries
`examples/agtau_list`	(a, g) and τ single-run scans
`examples/clump_sphere, amr_sphere, amr_tng, amr_ramses`	Clumpy and AMR media
`examples/dustemis`	Dust emission: Lucy+SEDUST, iteration, single-$T_{\rm eq}$, table, B&W, MRW
`examples/amr_sphere`	Dust emission on the AMR octree (`amr_dustemis.in`; matches the Cartesian sphere)
`examples/multipop`	Two stellar populations (hot young + cool old)
`examples/galaxy`	External galaxy SED (disk + bulge, edge-on / face-on)
`examples/milkyway`	All-sky map from an interior observer

Directory	Shows
`examples/benchmarks`	Camps (2015) SHG dust-emission benchmark and Mode 1/2 τ sweep

8. Validation

- **Regression.** The monochromatic scattering, the $(a, g)/\tau$ scans, and the clumpy-medium outputs remain bit-identical to v1.20 throughout the upgrade.
- **Dust-emission engine.** Run through the vendored SEDUST library with the Zubko et al. (2004) BARE-GR-S model, the emission spectra agree with the Camps et al. (2015) seven-code stochastic-heating benchmark to a median relative difference of 2–4% across $U = 10^{-2} \ldots 10^{4}$, and the emission peak matches the code median to within one wavelength bin.
- **Energy conservation.** The intrinsic dust SED integrates to the absorbed luminosity (to 10^{-5}); the two equilibrium methods (single-$T_{\rm eq}$ Lucy and Bjorkman & Wood) agree on both the absorbed fraction (to $< 0.3\%$ up to $\tau_V = 5$) and the dust temperatures.
- **AMR vs. Cartesian.** Dust emission on a uniform level-5 octree sphere reproduces the reprocessed fraction (0.863 at $\tau_V = 5$) and energy balance of the equivalent Cartesian sphere; Bjorkman & Wood and MRW on the octree conserve energy identically.

External three-dimensional transport benchmarks (DUSTY, Pascucci, the TRUST slab, and a direct Hyperion cross-check) require their published reference solutions and are added as they become available.

9. Algorithm Notes

Packet energy. Each packet carries a physical luminosity $L_{\rm pk}$ [erg/s]: $L_\star/N_\star$ for stellar packets and $L_{\rm dust}/N_{\rm dust}$ for dust-emission packets, so populations with different per-packet energies combine correctly in the J_λ tally.

Lucy iteration. The stellar J_λ is tallied once and reused; each iteration transports fresh dust-emission photons (sampled per cell from the emission spectrum), adds their J_λ to the stellar field, and recomputes the emission. The total emitted luminosity obeys $L^{(k)} = L^\star_{\rm abs} + f\,L^{(k-1)}$ with f the reabsorbed fraction, a geometric series converging to $L^\star_{\rm abs}/(1-f)$.

Bjorkman & Wood temperature. A cell's equilibrium temperature is found by inverting the monotonic $\kappa_B(T) = \int \kappa_{\rm abs}(\lambda)\,B_\lambda(T)\,d\lambda$ against the running absorbed power per H nucleus.

References

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